SUPERSTORE SALES ANALYSIS

Week 1 Report: Data Import, Power Query & Data Cleaning

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1. Problem Statement & Objectives:

A retail organization wants to analyze its sales data to understand customer purchases, product categories, regional performance, and profitability. Before creating a business intelligence dashboard, the raw Superstore dataset must be imported, cleaned, transformed, and prepared in Power BI. This report documents each step of that preparation.

Objectives:

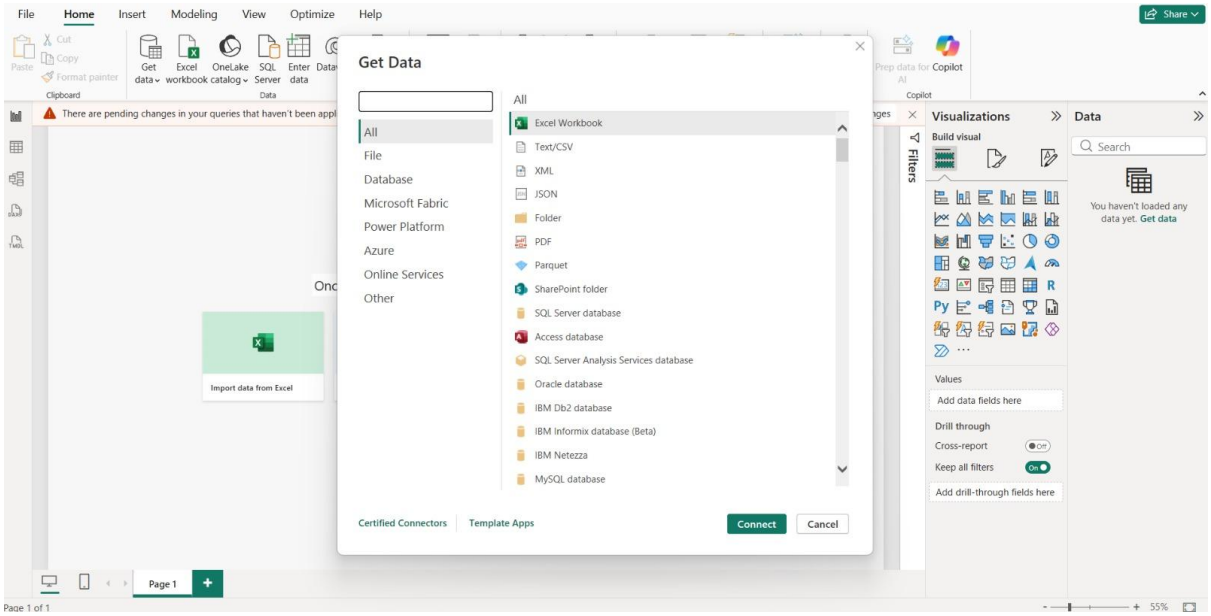
- Import data into Power BI Desktop and navigate Report, Data, and Model views.
- Inspect dataset attributes using Power Query Editor; verify data types, missing values, and record uniqueness.
- Perform column transformations (rename, split, text trimming, data type coercion).
- Extract temporal attributes from Order Date (Year, Quarter, Month, Day, Day of Week).

Tasks & Deliverables:

- Load Superstore CSV into Power Query Editor.
- Standardize text formatting across categorical attributes (Category, Sub-Category, Segment).
- Add custom date columns using Power Query M code / UI transformations.
- Close & Apply transformations to load a clean data model.

2. Getting Data into Power BI Desktop:

The first step is to bring the Superstore data into Power BI Desktop. In a new, empty report, the Get data option on the Home tab opens the Get Data window, which lists every supported source.

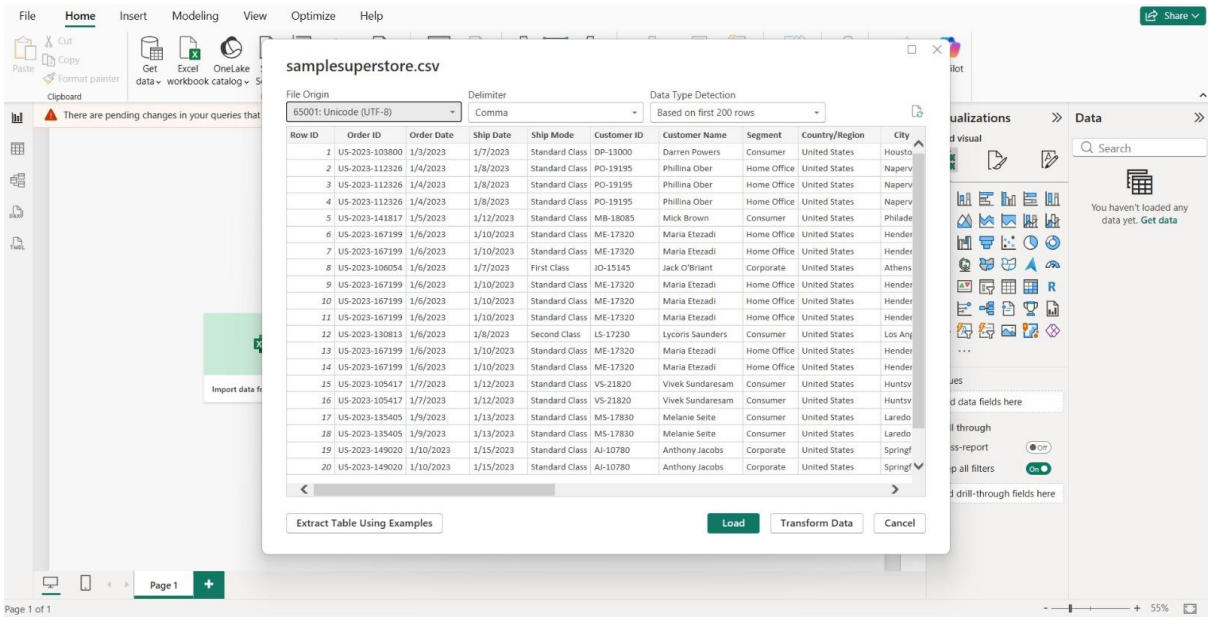


Key Points:

- **Source categories:** The window groups sources into All, File, Database, Microsoft Fabric, Power Platform, Azure, Online Services, and Other.
- **File connectors:** Excel Workbook, Text/CSV, XML, JSON, Folder, PDF, and several databases are available. Text/CSV is the connector for the Superstore CSV file.
- **Empty model:** The Data pane reads "You haven't loaded any data yet", confirming that this is the starting point of the workflow.

3. Previewing the Superstore CSV File:

After the CSV file is selected, Power BI shows a preview window so the file can be checked before it is loaded.

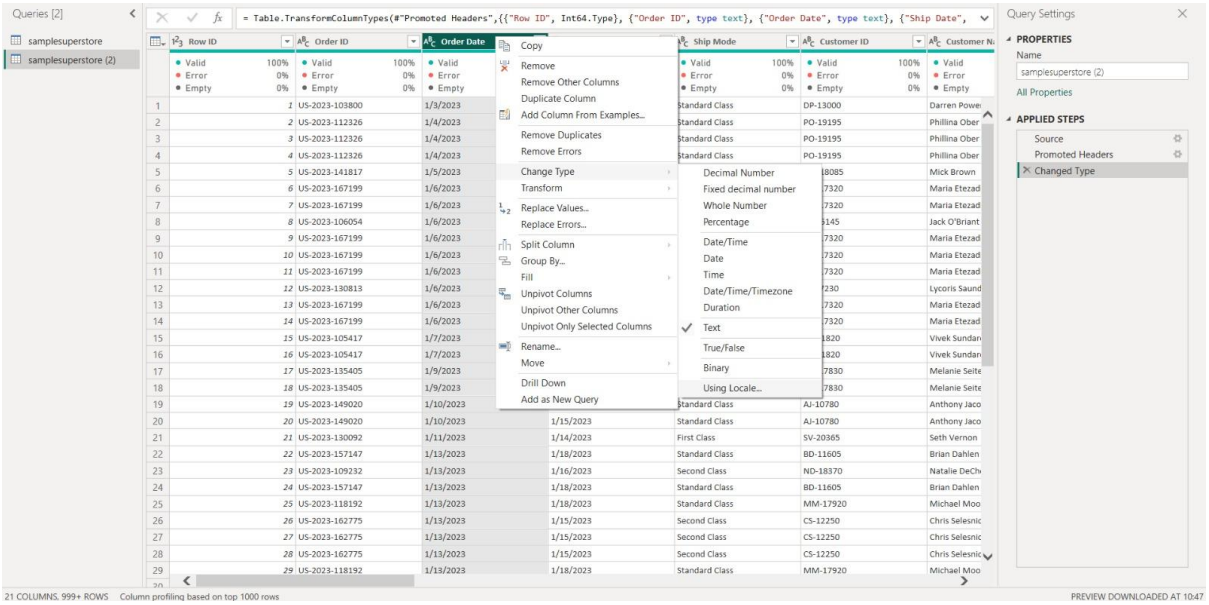


Key Points:

- **Import settings:** File Origin is 65001: Unicode (UTF-8), the Delimiter is Comma, and Data Type Detection is set to Based on first 200 rows.
- **Columns:** The preview shows Row ID, Order ID, Order Date, Ship Date, Ship Mode, Customer ID, Customer Name, Segment, Country/Region, City, and further columns.
- **Load vs. Transform:** Load brings the data in directly, whereas Transform Data opens Power Query Editor for cleaning. Transform Data is chosen so the data can be prepared first.
- **Date format:** Dates appear in month/day/year form (for example 1/3/2023), which matters for the date conversion step later.

4.Inspecting Data Types in Power Query Editor:

Power Query Editor is where the raw data is inspected and transformed. The query is shown with its Applied Steps on the right, and the column headers show the data type of each field.

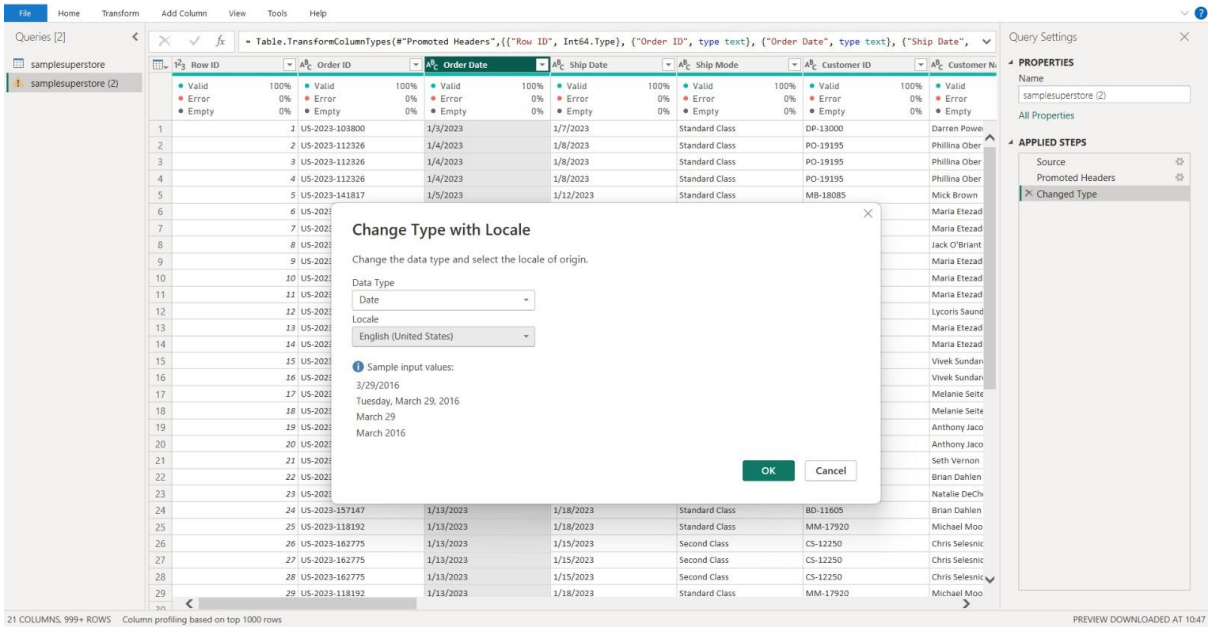


Key Points:

- **Initial steps:** The Applied Steps list shows Source, Promoted Headers, and Changed Type. The dataset has 21 columns and 999+ rows.
- **Data quality:** Column quality bars show 100% Valid, 0% Error, and 0% Empty for the visible columns, so no missing values or errors appear in the profiled rows (profiling is based on the top 1000 rows).
- **Data type issue:** Order Date is currently Text (shown by the ABC icon and the tick beside Text in the Change Type menu), so it must be converted to a Date type.
- **Chosen method:** Change Type > Using Locale is selected, because the text dates need to be interpreted with the correct regional format.

5. Converting Dates Using Locale:

The Change Type with Locale dialog converts a column while telling Power Query which regional format the original text uses.

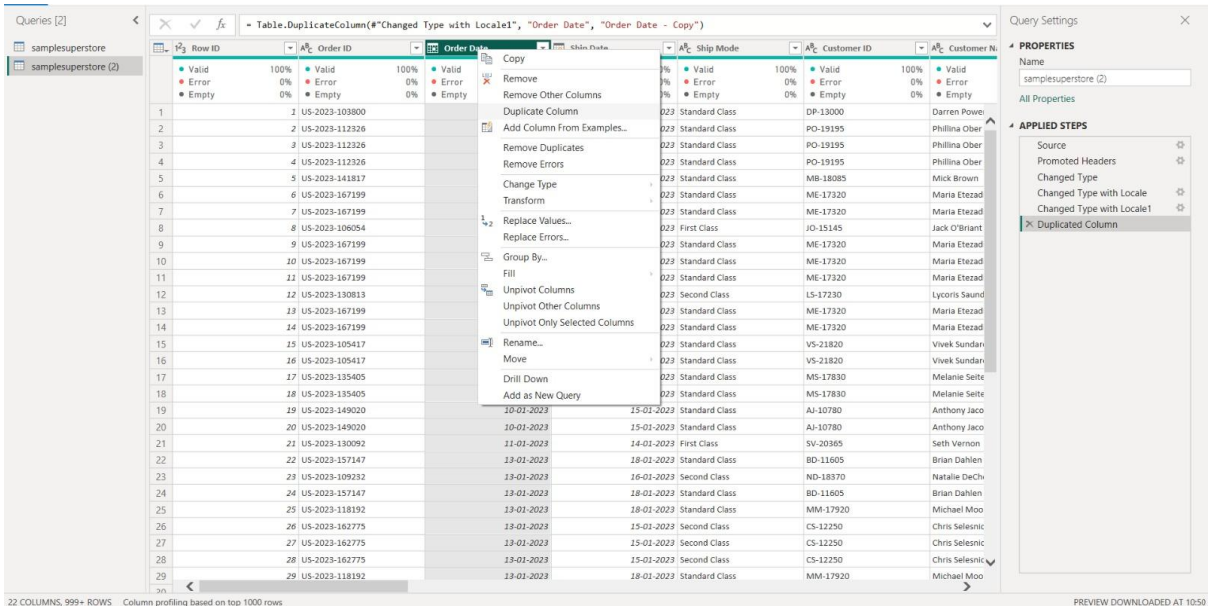


Key Points:

- **Settings:** Data Type is set to Date and Locale is set to English (United States), which matches source values such as 1/3/2023.
- **Sample values:** The dialog's sample input values (for example 3/29/2016 and Tuesday, March 29, 2016) confirm how the selected locale reads a date.
- **Result:** The same conversion is applied to Ship Date. Later screenshots show these steps recorded as Changed Type with Locale and Changed Type with Locale1.
- **Why it matters:** Choosing the locale avoids day and month being swapped, which would silently corrupt every date-based analysis.

6.Duplicating the Order Date Column:

To keep the original Order Date intact while creating date parts, the column is duplicated first.

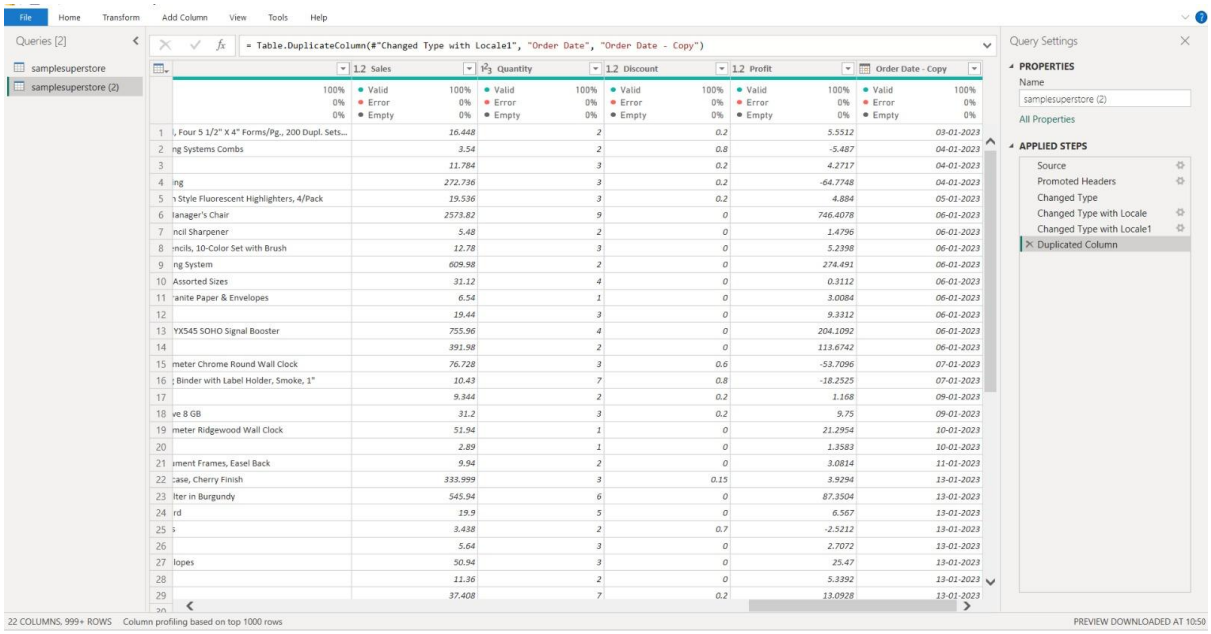


Key Points:

- **Action:** Right-clicking the Order Date header and choosing Duplicate Column creates an exact copy.
- **Applied Steps so far:** Source, Promoted Headers, Changed Type, Changed Type with Locale, and Changed Type with Locale1 have already been applied, so the dates are now true Date values (displayed as dd-mm-yyyy, for example 10-01-2023).
- **Benefit:** Working on a copy keeps the original column available for the date table and relationships in later weeks.

7. Verifying the New Column and Data Quality:

After duplication, the new column appears in the table and the step is recorded in Applied Steps.

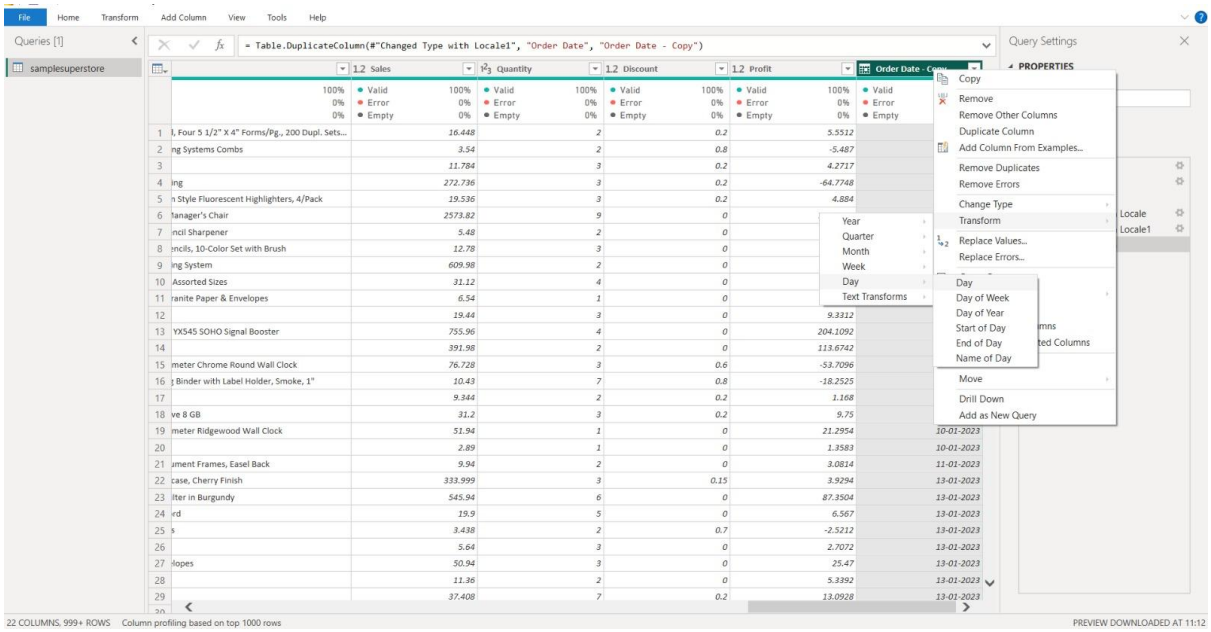


Key Points:

- **M code:** The formula bar shows the M code Table.DuplicateColumn(#"Changed Type with Locale1", "Order Date", "Order Date - Copy"), so the transformation is repeatable.
- **New column:** Order Date - Copy is now a Date column and the table has 22 columns (one more than before).
- **Data quality:** Sales, Quantity, Discount, Profit, and Order Date - Copy all show 100% Valid with 0% Error and 0% Empty in the column profile.
- **Queries:** The Queries pane lists samplesuperstore and samplesuperstore (2), so the original and working queries are both present.

8. Extracting Temporal Attributes:

The duplicated date column is used to derive the temporal attributes required for analysis. The Transform menu on the column provides ready-made date extractions.

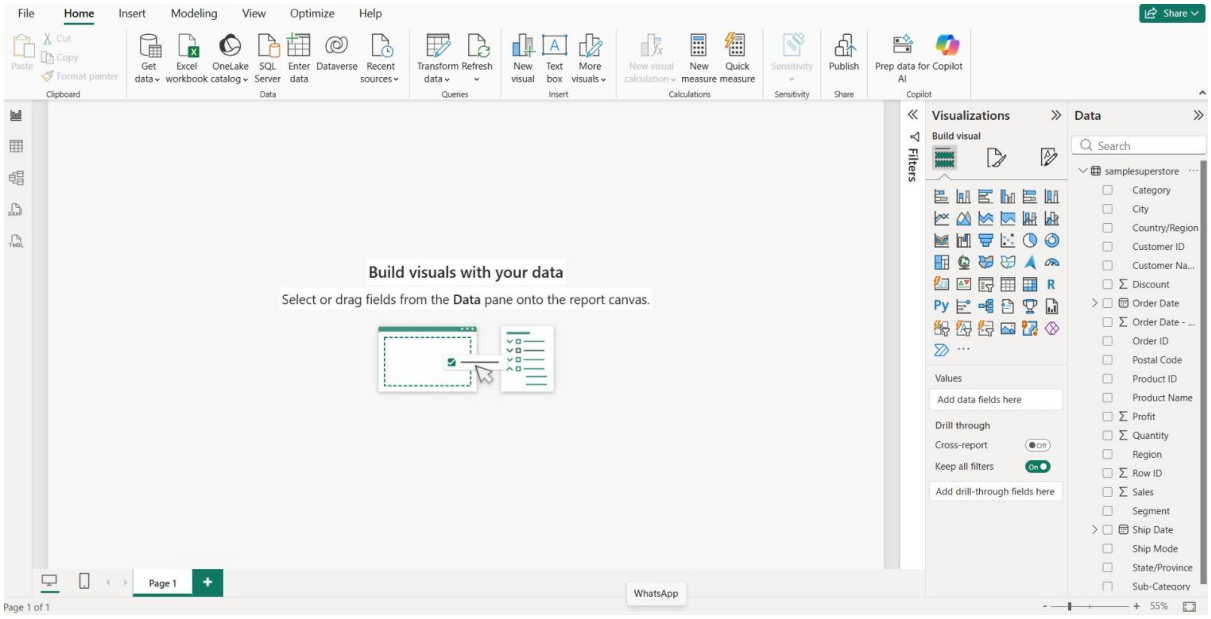


Key Points:

- **Date parts:** The menu offers Year, Quarter, Month, Week, and Day, which cover the required Year, Quarter, Month, and Day attributes.
- **Day options:** The Day submenu offers Day, Day of Week, Day of Year, Start of Day, End of Day, and Name of Day. Day of Week is the remaining required attribute.
- **Outcome:** Each extraction is recorded as an Applied Step, and the resulting numeric day value is what later drives the day-based line chart in the dashboard.

9. Close & Apply: The Loaded Data Model and Conclusion:

Once all transformations are complete, Close & Apply loads the cleaned query into the Power BI data model. The Report view then shows the loaded table in the Data pane, ready for modeling and visualization.



Key Points:

- **Loaded fields:** The Data pane lists samplesuperstore with fields such as Category, City, Country/Region, Customer ID, Customer Name, Discount, Order Date, Order ID, Postal Code, Product ID, Product Name, Profit, Quantity, Region, Row ID, Sales, Segment, Ship Date, Ship Mode, State/Province, and Sub-Category.
- **Views:** The report canvas is empty ("Build visuals with your data") and the left navigation gives access to the Report, Table, Model, and DAX query views.

Conclusion:

Week 1 took the raw Superstore CSV from import to a prepared data model: the file was previewed and loaded through Power Query, data types were verified and corrected using locale-aware conversion, additional date columns were created, and the transformations were applied. The dataset is now ready for the data modeling, DAX measures, and dashboard work in Week 2.